

Research Paper

Enhancing heat transfer from a circular cylinder undergoing vortex induced vibration based on reinforcement learning

Feng Ren^{a,*}, Fan Zhang^a, Yanan Zhu^b, Zhaokun Wang^c, Fuwang Zhao^c^a School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an, Shaanxi, China^b College of Chemistry and Chemical Engineering, Xi'an Shiyou University, Xi'an, Shaanxi, China^c Department of Mechanical Engineering, The Hong Kong Polytechnic University, Hong Kong, China

ARTICLE INFO

Keywords:

Active flow control
Thermal fluid–structure interaction
Reinforcement learning
Lattice Boltzmann method

ABSTRACT

Machine learning approaches have been proven to be effective and efficient in mediating flow dynamics so as to realize drag reduction, vibration suppression, etc. In this article, an intelligent active flow control approach is proposed to enhance heat transfer of a heated circular cylinder at $Re = 100$ undergoing vortex induced vibration in the lock-in condition. A high-fidelity solver based on the generalized interpolation-supplemented lattice Boltzmann method (GILBM) with moving reference frame (MRF) is used to establish the real-time numerical environment with structural motion, momentum transport, and heat transport. A reinforcement learning (RL) agent realized by proximal policy optimization (PPO) is adopted to establish the model-free controller, which receives sensorymotor cues as feedback signals and determines the real-time velocity issued from the jet as actuation. Through a group of trainings, each of whom involves 200 independent cases, the RL agent eventually learns an effective and efficient control strategy that improves the average heat transfer rate by 76.7%, quantified by the dimensionless Nusselt number. Through analyzing the spatiotemporal variations of vorticity and temperature fields, local Nusselt number and pressure coefficient along the cylinder surface, as well as the statistical flow quantities, it is proved that the enhanced heat transfer is contributed by both the mediated vortex formation and shedding in vicinity of the blowing jet, as well as the augmented vibration. Extra tests also confirm that the synthetic jets play a much more significant role than the augmented vibration. Furthermore, the success of the control strategy is attributed mainly by the phase lag between the maximum vibration displacement and the moment when the synthetic jets oppose their directions. Moreover, the more realistic but more complicated case with the non-Oberbeck-Boussinesq (NOB) approximation is also tested, confirming good self-adaptivity of the learnt strategy. Through this study, it is verified that the RL to be a powerful tool for mediating flows involving complicated momentum and heat transport.

1. Introduction

Flow past a bluff body is almost ubiquitous in nature and industrial field [1], which involves unsteady and asymmetrical loading, flow induced vibration (FIV), noise generation, heat transfer, etc. Researches on this topic have been quite hot, which can be found from many disciplines, such as fluid mechanics, thermodynamics, meteorology, etc. New knowledge gained from these persistent efforts also help improve performances of a lot of engineering equipment such as heat exchangers, power plants, riser systems in offshore platforms, etc. [2–4] For example, in chemical and pharmaceutical industries, the tube heat exchanger is a very typical type, which operates in the laminar or weakly turbulent regime and a high heat transfer rate is desired. Moreover, the heat exchanger is usually elastically mounted, and the effect of flow induced vibration shall not be neglected.

The most typical physical model is flow past a bluff body. In engineering applications, the bluff body is often elastically mounted, thus being allowed to oscillate due to unsteady fluid forces exerting on it. The flow induced vibration (FIV), being very rich in physical complexity, has attracted extensive attentions. According to various occurrence conditions and physical mechanisms, FIVs can be categorized into vortex induced vibration (VIV), galloping, flutter, buffeting, etc. In the last few decades, the VIV has been studied via theoretical, experimental, and computational approaches [5]. Owing to its generality, the classical physical model of VIV is adopted in this study to showcase the actively controlled momentum transport and heat transfer process.

Heat transfer accompanied by the structural vibration is also a significant issue. Regarding the forced oscillation scenario, Bao et al. [6] studied the effect of forced streamwise oscillations on heat transfer from a circular cylinder using the lattice Boltzmann method. Direct

* Corresponding author.

E-mail addresses: renfeng@nwpu.edu.cn (F. Ren), zhuyanana@xsyu.edu.cn (Y. Zhu), zhaokun.wang@connect.polyu.hk (Z. Wang).

evidences of thermal convection enhancement over the fixed situations are provided, especially when the oscillation amplitude is small and oscillation frequency is low. Sonawane et al. [7] studied the heat transfer characteristics of a circular cylinder that is forced to vibrate, where the Arbitrary Lagrangian–Eulerian formulation is used. Al-Mdallal and Mahfouz [8] further considered heat transfer from a non-rotating cylinder that is performing a circular motion, which is found to vary with Reynolds number, circular radius, and the motion frequency. This work revealed that heat transfer rate can be increased appreciably in the high range of above parameters.

In a more general FIV scenario, i.e., where the structural motions are coupled with the flow dynamics, the parametric trends and underlying mechanisms are different. Izadpanah et al. [9] studied comprehensively the VIV effects on the heat transfer from a circular cylinder at a Reynolds number of 150. Different combinations of structural stiffness and damping are investigated, which lead to different vibration mode such as the beating phenomenon. Sonawane et al. [10] performed cases studies of flow induced vibration of a cooled cylinder in a wide range of reduced velocity, where the effect of gravitational force is also taken into account. Ding et al. [11] studied heat transfer of a circular cylinder which is undergoing vibration induced by the wake of a upstream stationary cylinder. Results indicate that high reduced velocity can effectively strengthen heat transfer of the downstream cylinder. In the mean time, the wake induced vibration can also drastically enhance thermal convection of the upstream cylinder when the spacing is relatively small. In another configuration when two cylinders located in tandem are both elastically mounted, Sun et al. [12] studied effects of flow-induced vibration on heat transfer of the two cylinders in the laminar flow regime. Results show different phenomenon such as vibration responses, heat transfer rates, and wake patterns with different parameter conditions, revealing the rich and complex physical characteristics therein. Targeting at argument heat transfer, Kumar et al. [13] studied how to harness flow induced vibration of a D-section cylinder. It is then revealed that the VIV and galloping have significant influences on the thermal boundary layer, and further the heat transfer augmentation.

The aforementioned studies provide the qualitative and quantitative physical trends and underlying mechanisms of the FIVs associated with thermal convection. In realistic applications, enhancing the heat transfer is also of great interest. In the past decades, many active approaches have been proposed to control the flow dynamics especially flow past a bluff body [2,3]. The active flow control (AFC) approaches are then used to reduce drag, augment lift, mitigate vibration, and also enhance heating/cooling process. For instance, Arshad et al. [4] reviewed the synthetic jet (SJ) actuator used to enhance heat transfer in applications such as thermal management of electrical devices, and verified the SJ to be effective, energetically efficient, and reliable. With periodic rotary control, Ali et al. [14] studied the effect of rotational frequency on heat transfer of a circular cylinder, showing both augmentation and attenuation of heat transfer in certain parametric conditions. Particularly, this study also highlights the effect of non-Oberbeck–Bousinesq, which is more realistic. Heat transfer enhancement using rotary control is also evidenced in the experimental study of Ozalp et al. [15], where techniques such as particle image velocimetry (PIV) and thermal camera are used and provides consistent variation trend of heat transfer and fluid motion.

Note that the above AFC approaches are all open-loop. From the perspective of control theory, it is still doubtful whether they can work effectively when the environment is altered or interfered. In general, the closed-loop AFC is more promising in gaining good control performance while being robust if the environment is different. However, being very rich in physical complexity, the flow system makes it quite challenging in deriving an effective and efficient closed-loop controller. To the best of our knowledge, the research on such a controller for thermal-fluid–structure interaction system is still a blank area.

Machine learning has provided very promising solutions to address the complicated active flow control problems, as can be found in Refs. [16–18], as well as a more recent work by Vignon et al. [19]. To the best of our knowledge, there are two powerful machine learning tools that having been successfully applied to active flow control, i.e., the genetic programming (GP) and the reinforcement learning (RL). The GP models the control law as tree-like structures, which can be interpreted as symbolic expressions. Through the evolution process like the conventional genetic algorithm, the GP can optimize those symbolic expressions and finally provide the optimal control law [16,17]. Based on the GP, Ren et al. [20] investigated the closed-loop control strategy for suppressing the VIV of a circular cylinder in the lock-in condition. The RL models the control strategy using the neural network, whose parameters are optimized in an online manner and through trials and errors. Rabault et al. [21] applied the RL to reduce drag of a circular cylinder in a narrow channel by around 8%, where a pair of alternating blowing/suction jets are used to mediate the flow dynamics and overall 151 velocity sensors are used to provide feedback signals. With almost the same systematic configurations as those in Ref. [21], Ren et al. [22] further increased the Reynolds number by ten times to verify the effectiveness of the DRL in the more challenging turbulent environment. Varela et al. [23] applied the RL to exploit different physics through increasing the Reynolds number to 2000, and observed a new control strategy consisted of energization of the boundary layer and the separation zone. Ren et al. [24] also applied the RL to mitigate hydrodynamic wake signatures and realized hydrodynamic stealth. In this study, the learnt strategy was also extended to the VIV scenario. Zheng et al. [25] used the RL-guided synthetic jet control and suppressed the VIV by approximately 80%. Beintema et al. [26] applied the RL to control the Rayleigh–Bénard convection in the laminar regime with adjustable bottom heating. Recently, Guastoni et al. [27] applied the RL to a fully-developed and wall-confined turbulent flow case and outperforms the classical opposition control significantly. With these successful cases, machine learning approaches can potentially be applied for heat transfer enhancement with well designed closed-loop control system.

In this article, a simple yet efficient closed-loop active control approach is proposed to enhance heat transfer rate of a heated cylinder that undergoes VIV via the RL. In detail, the outline is organized as follows: in Section 2, the problem and the methodology are described, involving the reinforcement learning algorithm and the thermal-fluid–structure coupling solver; in Section 3, the RL exploration process, the exploitation of the optimal control strategy, as well as the effect of the NOB are demonstrated. In Section 4, conclusions are drawn. For the sake of continuity, the CFD validation and mesh convergence study are presented in Appendix.

2. Problem description and methodology

2.1. Physical problem description

In this study, the thermal convection around a circular cylinder are highlighted, which is immersed in liquid water and is elastically mounted. As depicted in Fig. 1, a cylinder with mass m and diameter D_0 is connected to a spring with stiffness of K_s . The upstream flow is uniform with a constant velocity of U_0 . The gravitational force is also neglected. At room temperature, i.e., $T = 293.15$ K, the fluid has a density of ρ_0 and a kinematic viscosity of ν_0 . The Reynolds number, defined by $Re = U_0 D_0 / \nu_0$, is fixed at $Re = 100$. Except for the cases discussed in Section 3.3, the Oberbeck–Bousinesq (OB) approximation is used, thus the density, kinematic viscosity, and thermal diffusivity α_0 of fluid are fixed. The Prandtl number, defined as $Pr = \nu_0 / \alpha_0$, is equal to 6.65, according to the property of liquid water and the formulae given in Ref. [28].

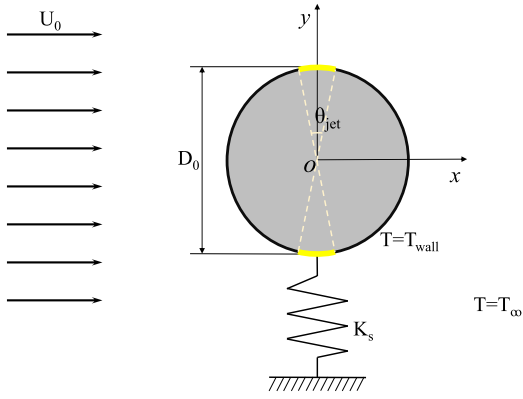


Fig. 1. Schematic diagram of the thermal fluid–structure interaction system.

The cross-flow motion of the cylinder is expressed as the following kinematic equation, i.e.,

$$m\ddot{y} = -K_s y + F_L, \quad (1)$$

where $\ddot{y}$ the instantaneous acceleration, y the displacement, and F_L the transverse force exerted on the cylinder by the surrounding fluid. In our simplified physical model, the structural damping is assumed to be zero. Two dimensionless parameters including the mass ratio and the reduced velocity, are then given as

$$m^* = \frac{m}{\rho_0 D_0^2}, \quad U_R = \frac{U_0}{f_N D_0}, \quad (2)$$

where $f_N = \sqrt{K_s/m}/2\pi$ is the natural frequency of this mass–spring system in vacuum. In this study, the parameter pair $m^* = 2$ and $U_R = 5$ are fixed. In this case, if the system is uncontrolled, the VIV will then falls into the lock-in condition [20], where the vibration amplitude is relatively high and the vortex shedding frequency is attracted to the natural frequency of the mass–spring system. Furthermore, the cylinder is heated and retains a constant temperature of $T_{wall} = 353.15$ K, thus providing a temperature difference of 60 K with the surrounding environment. With the upstream flow, the heat will then be convected downstream.

To enhance the heat transfer rate, a feedback control is utilized, where a pair of synthetic jets are used as the actuator (as shown in Fig. 1). The jets issue alternating blow/suction velocities such that the net flow flux is zero.

Local jet velocities are perpendicular to the cylinder surface, following a cosine distribution along the slots, which occupy an arc of 10° . This type of actuation has been used extensively in prior studies [21, 25], and our work [22] has also verified its effectiveness. Feedback signals are composed of the kinematic variables, i.e., the real-time $\{y, \dot{y}, \ddot{y}\}$, which can be obtained in experiments using the accelerometer. Compared to prior studies that utilize excessive velocity sensors both around the cylinder and in the wake [21,22,25], our choice is much simpler and more realistic. The controller is then formulated as

$$u_{jet} = f(y, \dot{y}, \ddot{y}), \quad (3)$$

which is determined by the reinforcement learning agent introduced in Section 2.3. In order to focus on enhancing the heat transfer efficiency, we choose not to consider factors such as the energy consumption and potential increase in hydrodynamic forces exerted by the surrounding fluid to the cylinder. The following reward function is then defined to evaluate the control performance, i.e.,

$$r = \langle Nu \rangle, \quad (4)$$

where Nu denotes the instantaneous Nusselt number of the cylinder, which is an integral of the local temperature gradient along the cylinder

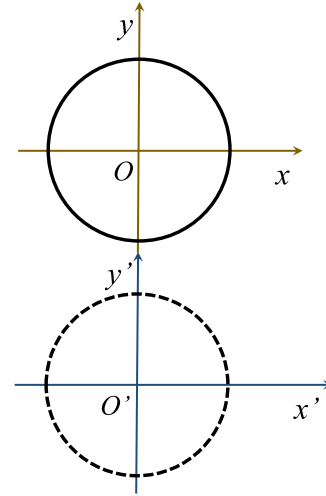


Fig. 2. Schematic diagram of the coordinate transformation.

surface, i.e., $Nu = -\frac{D_0}{2\pi\Delta T} \int_0^{2\pi} \frac{\partial T}{\partial r} d\theta$. Here ΔT denotes the temperature difference between the cylinder surface and the environment.

For the sake of generality, the length scales in the remainder parts are given as normalized values by D_0 . The velocity is normalized by U_0 , time scale normalized by $T_0 = D_0/U_0$, pressure normalized by $\frac{1}{2}\rho_0 U_0^2$, and forces normalized by $\frac{1}{2}\rho_0 U_0^2 D_0$.

2.2. Numerical solver for the thermal-fluid-structure coupling system

2.2.1. Governing equations for moving reference frame

The most common practice of the CFD is performed in an absolute coordinate system. For this case, treatment of moving boundaries is usually difficult and might brought extra errors. Another alternative is to consider the moving frame, where the Navier Stokes equation (NSE) in the dynamic coordinate system is computed. In this study, the latter practice is chosen owing to its advantageous accuracy in near-wall flow dynamics and the relatively simplicity. With only the unsteady translation along transverse direction being considered, the coordinate transformation is stretched in Fig. 2, where $O'x'y'$ represents the absolute reference frame (AFR) and Oxy the moving reference frame (MFR).

Following the work of Li et al. [29], the governing incompressible NSE in the MFR can be formulated as

$$\nabla \cdot \mathbf{v} = 0$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho_0} \nabla p + \nu_0 \nabla^2 \mathbf{v} + \mathbf{F}_{MFR}, \quad (5)$$

where $\mathbf{F}_{MFR} = (0, -\ddot{y})$ is the extra body force that models the effect of unsteady translation. In addition, the far-field boundary condition is also affected by the coordinate transformation, which is then modified as

$$\mathbf{v}(\mathbf{x}, t) = (U_\infty, -\ddot{y}). \quad (6)$$

Similar as the way Li et al. [29] derives the NSE in the MFR, one can also derive the heat transport equation therein. First, the heat transport equation in the AFR is written as

$$\left(\frac{\partial \phi'}{\partial t} \right)_a + (\mathbf{v}' \cdot \nabla_a) \phi' = \alpha \nabla_a^2 \phi', \quad (7)$$

where the subscript 'a' corresponds to the AFR. ϕ' denotes the temperature in the AFR, and $\phi = \phi'$. The temporal term can then be transformed as

$$\left(\frac{\partial \phi'}{\partial t} \right)_a = -(\dot{y} \cdot \nabla) \phi + \left(\frac{\partial \phi}{\partial t} \right), \quad (8)$$

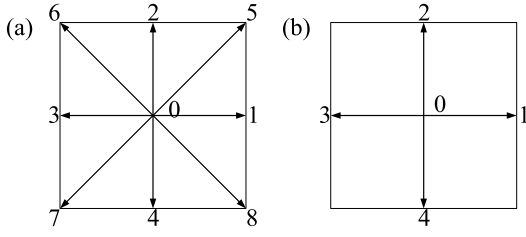


Fig. 3. Schematics of the lattice Boltzmann discretization scheme (a) D2Q9 and (b) D2Q5.

the convection term as

$$(\mathbf{v}' \cdot \nabla_a) \phi' = [(\mathbf{v} + \mathbf{j}) \cdot \nabla] \phi, \quad (9)$$

and the diffusion term as

$$\alpha \nabla_a^2 \phi' = \alpha \nabla_r^2 \phi. \quad (10)$$

Substituting Eqs. (8), (9) and (10) into Eq. (7), one can obtain the mathematical expression of the heat transport equation in the MFR, i.e.,

$$\frac{\partial \phi}{\partial t} + (\mathbf{v} \cdot \nabla) \phi = \alpha \nabla^2 \phi. \quad (11)$$

It can be noted from Eq. (10) that the heat transport equation after coordinate transformation still retains the form of a convection-diffusion equation (CDE), which can be easily solved under the framework of the LBM.

2.2.2. LBE for momentum transport

In order to obtain the spatiotemporal evolution of the flow dynamics in the MFR, the lattice Boltzmann method (LBM) is applied for the momentum transport. First, the general formula of the lattice Boltzmann equation (LBE) for the momentum transport is expressed as [30]

$$\mathbf{f}(\mathbf{x} + \mathbf{e}_{f,i} \delta t, t + \delta t) = \mathbf{f}(\mathbf{x}, t) - \mathbf{M}_f^{-1} \mathbf{A}_f \mathbf{M}_f [\mathbf{f}(\mathbf{x}, t) - \mathbf{f}^{eq}(\mathbf{x}, t)] + \mathbf{M}_f^{-1} (\mathbf{I} - \frac{1}{2} \mathbf{A}_f) \mathbf{M}_f \mathbf{S} \delta t \quad (12)$$

where $\mathbf{f}$ is the particle distribution function. The multi-relaxation time (MRT) scheme [31] is adopted to improve numerical stability. Herein $\mathbf{M}_f$ is the orthogonal matrix that transforms the particle distribution function from velocity space to the moment space, $\mathbf{I}$ the unit matrix, $\mathbf{A}_f$ the diagonal matrix that contains the relaxation times. δx and δt represent one lattice unit length and one lattice time step, respectively. The two-dimension and nine-speed (D2Q9) scheme is adopted, which means the model is two-dimensional and is discretized in nine directions, as shown in Fig. 3(a). The particle streaming direction is given as

$$\mathbf{e}_f = c \begin{bmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{bmatrix}. \quad (13)$$

Weighting factors for the D2Q9 scheme are $w_0 = 4/9$, $w_{1 \sim 4} = 1/9$ and $w_{5 \sim 8} = 1/36$. The diagonal matrix that contains the relaxation times is set as $\mathbf{A}^g = \text{diag}[1.0, 1.1, 1.0, 1.0, 1.2, 1.0, 1.2, s_v, s_v]$, where the kinematic viscosity is related to s_v via $s_v^{-1} = 0.5 + \nu_{0e}/c_s^2 \delta t$. The transformation matrix is given as

$$\mathbf{M}_f = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -4 & -1 & -1 & -1 & -1 & 2 & 2 & 2 & 2 \\ 4 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & -2 & 0 & 2 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \\ 0 & 0 & -2 & 0 & 2 & 1 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \end{bmatrix}. \quad (14)$$

Based on the incompressible model proposed by He and Luo [32], the i th component of the equilibrium function f_i^{eq} is expressed as

$$f_i^{eq} = w_i \frac{\rho}{c_s^2} + w_i \rho_0 \left[\frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right], \quad (15)$$

where $c_s = c/\sqrt{3}$ is lattice sound speed, $c = \delta x/\delta t$ is lattice speed. $\rho_0 = 1 \text{ mu} \cdot \text{lu}^{-3}$ denotes the fluid density. Following the work of Guo et al. [33], the source term $\mathbf{S}$ that models the effect of body force in Eq. (12) is written as

$$\mathbf{S}_i = w_i \rho_0 \left[\frac{\mathbf{e}_i \cdot \mathbf{F}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})(\mathbf{e}_i \cdot \mathbf{F})}{c_s^4} - \frac{\mathbf{u} \cdot \mathbf{F}}{c_s^2} \right]. \quad (16)$$

Then the local macroscopic variables can be obtained with the distribution functions, i.e.,

$$\rho = \sum f_i c_s^2, \quad \mathbf{u} = \frac{1}{\rho_0} \left(\sum f_i \mathbf{e}_i + \frac{1}{2} \mathbf{F} \right). \quad (17)$$

2.2.3. LBE for heat transport

The LBM is also used to solve the heat transport in the MFR, as formulated in Eq. (10). Like the LBE for momentum equation, the general LBE with the MRT form [34,35] is firstly presented, i.e.,

$$\mathbf{g}(\mathbf{x} + \mathbf{e}_{g,i} \delta t, t + \delta t) = \mathbf{g}(\mathbf{x}, t) - \mathbf{M}_g^{-1} \mathbf{A}_g \mathbf{M}_g [\mathbf{g}(\mathbf{x}, t) - \mathbf{g}^{eq}(\mathbf{x}, t)] \quad (18)$$

where $\mathbf{g}$ represents the particle distribution functions, $\mathbf{M}_g$ the transformation matrix, and $\mathbf{A}_g$ the diagonal relaxation matrix. In terms of the spatial discretization scheme, the two-dimension and five-speed (D2Q5) scheme [36] is adopted, as shown in Fig. 3(b). Then the particle streaming direction is given as

$$\mathbf{e}_g = c \begin{bmatrix} 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix}. \quad (19)$$

The weighting factors in the D2Q5 scheme are $w_0 = 1/3$, and $w_{1 \sim 4} = 1/6$. The diagonal matrix that contains the relaxation times is set as $\mathbf{A}_g = \text{diag}[1.0, s_a, s_a, 1.2, 1.0]$, where the thermal diffusivity is related to s_a via $s_a^{-1} = 0.5 + \alpha_{0e}/c_s^2 \delta t$. The transformation matrix is given as [36]

$$\mathbf{M}_g = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ -4 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 & -1 \end{bmatrix}. \quad (20)$$

The equilibrium function is expressed as

$$g_i^{eq} = w_i \phi \left(1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} \right). \quad (21)$$

The local order parameter for temperature is then solved via

$$\phi = \sum g_i, \quad (22)$$

where ϕ is a parameter varying linearly with temperature. Here $\phi = 0$ and 1 correspond to temperature T_∞ and T_{wall} , respectively.

2.2.4. Generalized interpolation-supplemented LBM

The conventional LBM usually uses uniform Cartesian mesh, thus simulations of flow past a bluff body such as a circular cylinder relies much on the curved boundary treatment, including mainly the interpolation-based method, and the immersed boundary method [37]. The former uses various interpolation methods to reconstruct the unknown post-collision particle distribution functions with the surrounding known ones, so as to enforce the no-slip boundary condition, while unavoidably introducing unfavorable effects to mass and momentum conservation, as well as stress continuity. For the latter, the Lagrangian mesh that models the geometrical shape of the bluff body is introduced, and the boundary condition is transformed into body forces of neighboring Eulerian nodes, which introduces extra numerical diffusion to

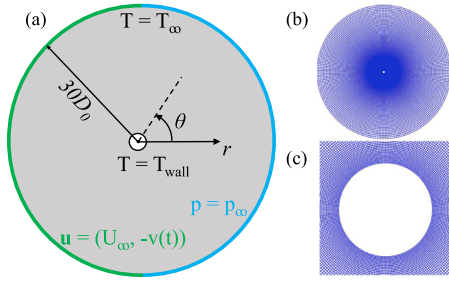


Fig. 4. Schematics of (a) the computational domain, (b) the whole mesh partition and (c) the enlarged near-wall mesh setup.

the boundary layer. In this study, we follow the generalized form of interpolation-supplemented LBM (GILBM), where the particle collision processes for momentum and heat transport are unaffected. The only modification lies in the streaming step, where the physical plane is transformed into the computational plane. Here the computational domain uses uniform Cartesian mesh identical with that of the conventional LBM, while the computational domain can use nonuniform mesh to gain better accuracy especially in the near-wall zone. Detailed introductions of the GILBM can be found from Ref. [38–40]. Fig. 4 shows the computational domain and mesh setups.

In the simulations, the azimuthal meshes are uniformly partitioned, and the radial meshes are distributed exponentially. The mathematical expressions can be written as

$$\theta_i = \frac{2\pi x_i}{NX}, \quad r_j = \frac{D_0}{2} \exp\left(\frac{\alpha y_j}{NY-1}\right), \quad (23)$$

where $\alpha = \ln(2r_{max}/D_0)$. $r_{max} = 30D_0$ is the maximum radius. The node index in the computational domain is $x_i = 0, 1, 2, \dots, NX-1$ and $y_j = 0, 1, 2, \dots, NY-1$.

Boundary conditions are sketched in Fig. 4(a). At the cylinder wall, no-slip and no-penetration velocity boundary is enforced. At the outer boundary, the velocity specified as $(U_\infty, -v(t))$ is enforced at the left half of the physical domain, while the right half is enforced with constant pressure $p_\infty = 0$. In addition, constant temperature T_{wall} is enforced at the cylinder wall, while T_∞ at the outer boundary. These boundary conditions are realized by the non-equilibrium extrapolation scheme [41].

To calculate the kinematic equation of Eq. (1), the lift force is calculate by integrating pressure and shear stress along the cylinder surface. In this step, the local shear stress is solved with tangential velocity of the first three mesh layers using a third-order finite difference approach. By the way, the calculation of local Nusselt number also follows the same differential scheme. After obtaining the instantaneous lift force, the kinematic equation is solved by the second-order Eulerian scheme.

The above in-house thermal fluid–structure interaction solver is accelerated with GPU parallel computing technique, and is performed with an Intel Xeon E5-2678 CPU and an NVIDIA RTX 3090 GPU. This efficient solver is able to complete one training case within 4 min.

2.3. Reinforcement learning based active flow control

The RL is used to obtain effective control strategies for enhancing heat transfer rate. The AFC framework is sketched in Fig. 5, where the RL agent interacts with the thermal fluid–structure interaction environment in a real-time manner. The RL agent receives state information consisting of the sensorymotor cues, and determines the action via the policy network. Meanwhile, from the policy-based RL algorithms, the RL agent also needs to gain the ability of predicting the long-term control performance with the current state information. To realize this, the instantaneous reward that estimates the control

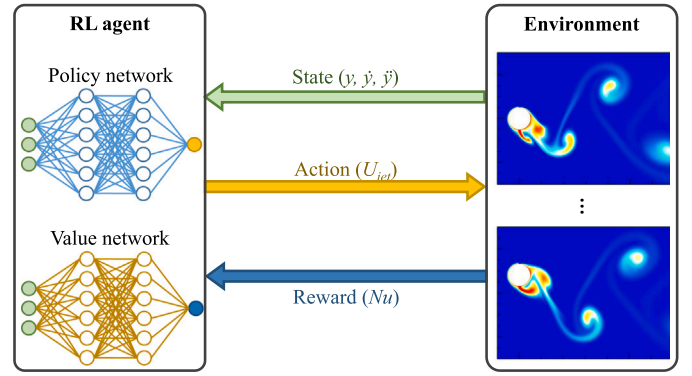


Fig. 5. Schematic diagram of the interaction between the RL agent and the environment.

performance is provided by the environment and is used to train the value network. The input of both networks is the instantaneous state information, while the policy network outputs the action and the value network outputs the predicted long-term performance. The schematic diagram of the control system is shown in Fig. 5. Here the state of the environment is represented by the sensorymotor cues, and the action is the velocity issued by the synthetic jets operating in the one-blowing-one-suction mode. The reward defined in Eq. (4) is then evaluated and fed to the agent, providing a baseline for the agent to learn how to evaluate the control performance, so as to encourage the agent to learn a better controller that achieves higher reward.

The proximal policy optimization (PPO) [42] is applied to realize the RL agent, which is advantageous in determining continuous actions. In the present study, our in-house PPO code is developed under the framework of TensorFlow, wherein the policy network and value network are realized using the multi-layer perceptron (MLP) with two hidden layers, each having 60 neurons. In the very beginning, both networks are initialized with random weights and biases, indicating that the RL agent does not rely on any prior knowledge of this non-linear and high-dimensional system. Moreover, the RL agent also begin to interact with the environment with randomized actions. In each episode, a DRL agent runs the policy and collects a trajectory, i.e., a sequence of state–action–reward combinations, i.e., $\tau = (s_1, a_1, r_1), (s_2, a_2, r_2), (s_t, a_t, r_t), \dots, (s_N, a_N, r_N)$. To consider a long-term effect of the actions, a discounted reward $R_t = \sum_{t'=t}^T \gamma^{t'-t} r_{t'}$ is used, where $\gamma = 0.999$ is the discount factor. This definition gives later rewards more weights in calculating R_t . While exploring the environment through stochastic trials and errors, the agent gradually learns how to exert a certain action when the environment is at a certain state through the policy network, as well as how to evaluate the long-term control performance through the value network. By using this PPO-based framework, it is expected that the agent can learn effective control strategies through stochastic trials and errors. More details about this framework and the associated hyper-parameters can be found from our prior studies [22,24].

Like many machine learning applications that generally involve an exploration–exploitation process, the RL-based AFC can also be divided into the training stage and the deterministic control stage. In the very beginning of the training stage, parameters of the policy network and the value network are randomly initialized, indicating that the RL agent does not have any prior knowledge about this thermal fluid–structure interaction system. The training consists of a designated number of episodes (200 in this study), where in each episode, the real-time environment is reinitialized with fully developed flow and thermal field, as well as the structure motion. In each episode, the CFD simulation lasts for $120T_0$, approximately 22 vortex shedding periods of the uncontrolled case (as can be seen from Table 5). States are extracted 10 times within one T_0 , which is used to determine the real-time action in the same

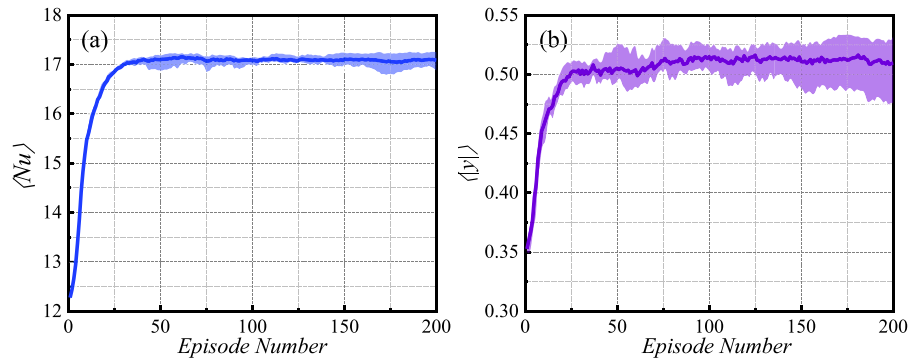


Fig. 6. Variations of time-averaged (a) Nusselt number and (b) the cross-flow vibration displacement during the learning process.

Table 1

Hyper-parameters used in the RL during the training stage.

Parameter	Value/Method
Network architecture (hidden layer)	Two 60×60 fully-connected layers
Actuations per T_0	10
Length of each episode	$120T_0$
Optimizer	Adam [44]
Action distribution	Beta [43]
Discount factor (actor & critic)	0.97
Learning rate (actor)	Exponentially decaying from 10^{-3} to 10^{-4}
Learning rate (critic)	Exponentially decaying from 10^{-3} to 10^{-4}

pace. Then the instantaneous reward is calculated accordingly. After completing one CFD case, the state–action–reward chain obtained from one whole episode is then provided to the RL agent so as to update the neural networks to the next episode. The hyper-parameters used in this study are listed in Table 1. Here the adaptive moment estimation (abbreviated as “Adam”) optimizer is used when updating parameters of both neural networks, where relatively small learning rates are used to make the training converge more stably. To enlarge the search space, actions are sampled from the bias-free Beta distribution [43].

3. Results and discussions

3.1. Exploring control strategy

Five independent training trials are performed where the neural networks are initialized with different random seeds so that one can reduce the influence brought by the randomness during the learning, such as the randomness in action sampling. The learning curves are shown in Fig. 6, where two time-averaged quantities within each episode are presented, i.e., Nu and $|y|$, to quantify the variations of heat transfer rate and vibration amplitude. In this figure, the semitransparent bands represent the upper bound and lower bound of all five trials, and the bold lines show the results averaged by these trials.

As can be seen from Fig. 6(a), from the very beginning to the end, the learning proceeds with a rapid increasing stage and a slowly converging stage. The former involves around 30 episodes, during which the averaged Nusselt number is increased by almost 40%, which is the first direct evidence that the RL agent gradually learns an effective control strategy of enhancing heat transfer rate. Meanwhile, as in Fig. 6(b), the VIV amplitude quantified by $|y|$ follows a similar trend, and eventually reaches approximately 1.4 times that of the starting point. This similar trend of Nu and $|y|$ implies the close relationship between the enhanced heat transfer and augmented vibration. This relationship can be easily understood because the augmented vibration could usually prompt fluid mixing between the near-wall hot fluids and the surrounding cold fluids, as evidenced by some prior studies.

However, comparing Figs. 6(a) and 6(b), one can observe another very interesting fact. In the converging stage of the RL learning, only

very slight fluctuations can be found from $\langle Nu \rangle$, suggesting that all five trials leads to very close control performance. By contrast, the variation of $\langle |y| \rangle$ becomes volatile in this stage. This fact implies that the enhanced heat transfer is not only associated with the augmented vibration, and there must be other factors that are attributed to the heat transfer enhancement. This issue will be figured out in Section 3.2.

3.2. Exploiting the optimal control strategy

After the trainings reach convergence, the optimal control strategy is obtained, modeled by the policy network. The deterministic control is then carried out to exploit the optimal control strategy, where each actuation is determined as the mathematical expectation of the action distribution mentioned in Section 3.1, thus involving no randomness. The CFD simulation now lasts for $200T_0$, and the actuations are updated more frequently (100 times in one T_0) so as to perform smoother actuations. From this section on, simulations are carried out using the fine mesh to provide more reliable data.

Comparisons of temporal variations of global quantities including the Nusselt number Nu , the VIV displacement y , and the velocity issued from the jet u_{jet} are depicted in Fig. 7. The AFC is turned on at $t = 0$, when the cylinder is located near its equilibrium point. Thereafter, the u_{jet} varies periodically and shows a rectangle-like wave form, where the upper and lower limits correspond exactly to the constraint enforced. The immediately responding forcing makes Nu and y falls into a quasi-steady state after only about one vibration cycle. In this quasi-steady stage, dramatic increase of Nu is noted, where the mean Nu becomes 21.6, a 76.7% increment after control. This direct evidence confirms again the successful heat transfer enhancement. Like the trend shown in Fig. 6(b), the change in Nu is also accompanied by the augmented vibration, where the amplitude of y is increased by $\times x$ after control. Furthermore, Fig. 7(b) also shows a significant frequency shift after control, which is 0.183 before control and 0.174 after control. As a matter of fact, the vibration frequency coincides with the vortex shedding frequency in the quasi-steady state, which also matches the control frequency in the controlled case. In the uncontrolled case, the VIV is in the lock-in state, where the vibration cycle and vortex shedding cycle is determined by the mass–spring system. The frequency shift corresponds to the delayed vortex formation and shedding, and it is viewed as a significant effect of the AFC.

To gain a clearer view, temporal variations of y , Nu , and u_{jet} within one vibration cycle of the quasi-steady state are illustrated in Fig. 8. The time axis is now modified as $t' = (t - t_E)/T_{vib}$, where t_E represents the instant when $y = 0$ and $\dot{y} > 0$, and T_{vib} denotes the vibration period. This simple modification allows a direct comparison of flow characteristics at the same vibration phase. With that, one can notice an interesting phenomena from Fig. 8(c) that the instant when U_{jet} turns sharply from the negative peak to the positive peak corresponds to neither when y reaches the maxima nor when $\dot{y}$ reaches its peak. The phase lag being 14.5° in the first half cycle and 20.0° in the second half is noticed and

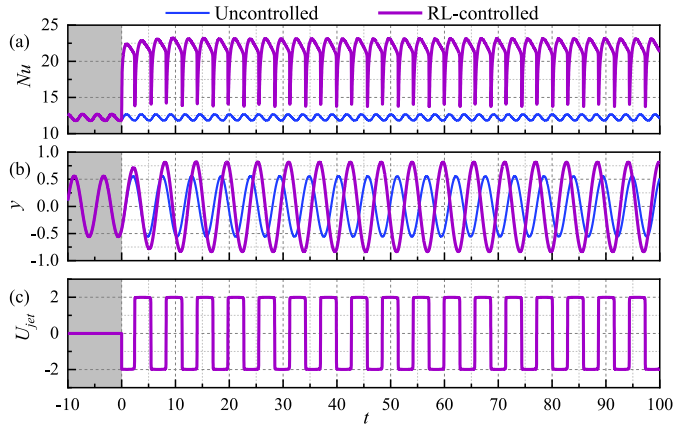


Fig. 7. Temporal variations of global quantities in the deterministic control, including (a) Nusselt number, (b) transverse displacement, and (c) velocity issued from the jet.

marked in Fig. 8(c). From a physical point of view, when $t' \in [0, 0.25]$, the cylinder moves upward and reaches the maximum displacement. At this instant, the control strategy does not make any change, but holds a while so as to encourage larger upward motion, until it can no longer balance the elastic force exerted by the spring. Then U_{jet} opposes its direction to accelerate the downward motion. Therefore, the phase lag is the key to the success of the RL-based AFC, and also the reason for shifted frequency observed in Fig. 7. Furthermore, the difference in $\Delta\phi_1$ and $\Delta\phi_2$ explains the slight asymmetry in Fig. 8(b). As a matter of fact, even well converged, the RL would usually lead to imperfect strategies. However, the marginal asymmetry here is acceptable in practical applications.

To better understand the contribution of the synthetic jets, we proceed with qualitative aspects of the spatiotemporal evolution of vortical structures and temperature distribution, as in Figs. 9(a–d). Three instants are chosen, where $t' = 0, 1/8$, and $1/4$. The first column of Fig. 8 corresponds to the moment when $y = 0$ and $\dot{y}$ reaches its maxima, while the third column corresponds the moment when y reaches its maxima and $\dot{y} = 0$. Due to the periodicity of y in temporal domain, the two instants are regarded as the most representative ones. Figs. 9(a–b) show results of the uncontrolled case, and Figs. 9(c–d) correspond to the controlled case. Besides, quantitative results of local Nusselt number distributed along the cylinder surface is given in Fig. 9(e). In subfigures (b) and (d), vortical structures are marked by the λ_{ci} criterion [45], together with the vorticity contour.

As can be observed from Figs. 9(b) and 9(d), in both cases, two single (S) vortices are shed within each vibration cycle, while being different from the more common 2S mode in that the vortices would coalescence in the wake. Therefore, the vortex shedding patterns are then identified as the C(2S) mode [5]. The C(2S) mode is quite common in VIV, especially in the resonance region of interest. For the C(2S) mode, the vibration amplitude is usually small, and the wake pattern would possibly become the C(P+S) mode if the vibration amplitude was further augmented [5]. However, this transition is not observed in this study, indicating that the flow stability and vortex dynamics is not essentially altered after control.

Similar patterns of the temperature field and the vorticity field also indicates their very close relationship. An overall view can be drawn from Fig. 9 is the higher temperature distributed along with the strengthened vortical structures, which is also an evidence that more heat is transported from the cylinder surface and advected downstream. Comparing Figs. 9(a) and 9(c), one can observe another significant phenomenon that a lot of heat is released from the blowing jet slot, which rolls up with the upstream flow and mixes with the shear layer formed at the upper surface of the cylinder. By contrast, the temperature field near the suction slot shows little difference after

control. Quantitative results in Fig. 9(e) support the above qualitative perspective. In the RL-controlled case, the local Nusselt number at the blowing slot as well as its rear shows very remarkable enhancement of heat transfer rate. Meanwhile, at the suction slot, one can note the negative role of the synthetic jet while its surrounding fluids being forced to convect to the cylinder surface also enhances the local heat transfer. Furthermore, as shown in Fig. 8, the fact that Nu is slowing decreasing while U_{jet} is fixed indicates that the vibration does have an impact role in the final control performance. Therefore, Fig. 9 reveals the changes in flow and temperature field before and after control, and provides a direct evidence that the significantly enhanced heat occurs in vicinity of the blowing jet.

Time averaged distributions of Nu and pressure coefficient C_p along the cylinder surface are then depicted in Fig. 10. Beginning with Nu , according to the difference between the uncontrolled and controlled data, the whole distribution is divided into three zones. Herein Zone I is near the front stagnation point, where the local Nu is almost unaffected by the AFC. In Zone II, dramatic changes occur after control as evidenced in Fig. 9(e). Zone III correspond to the zone near the rear stagnation point, where the thermal convection efficiency is slightly mitigated. Because this zone is too small, its impact on the overall heat transfer rate is only marginal. It can then confirmed that Zone II contributes most to the final enhanced heat transfer, which is in vicinity of the synthetic jets. The effect of AFC on the C_p distribution, as in Fig. 10(b), shows a quite different trend from Fig. 10(a). Here the jet control causes impacts on all three zones, especially the sharp pressure drop at the jet slot, suggesting that the pressure is more sensitive to actuations normal to the wall. With evidences in Figs. 9 and 10, it is verified that the blowing jet plays the dominating role in the successful AFC. From a physical perspective, the blowing velocity issued to the surrounding environment prompts the mixing of near-wall hot fluids with the surrounding cool fluids. In th mean time, the vortex formation and shedding process are significantly affected, leading to different distributions of normal wall stress and transverse lift force. Eventually the vibration is augmented and facilitates heat transfer in turn.

To elucidate the physical rationale of the enhanced heat transfer from the aspect of flow and thermal fields. Time-averaged first-order and second-order statistical quantities related to the flow and thermal quantities are illustrated in Fig. 11. Here the $\langle \cdot \rangle$ operation means that the quantity inside is averaged in temporal space. The subscript ' means the fluctuation part, for example, the instantaneous ϕ is decomposed as the mean part $\langle \phi \rangle$ and the fluctuation part ϕ' . Besides, a quantity such as $\langle u'\phi' \rangle$ is calculate as $\langle u\phi \rangle - \langle u \rangle \langle \phi \rangle$. When performing the time-averaging, in order to avoid the influence of the vibrating body, part of the whole domain are selected, which ranges from $x = 0.52$ to $x = 10$ and from $y = -3$ to $y = 3$. This subdomain roughly covers the wake region of interest. Significant changes are observed from all quantities in Fig. 11. Starting from $\langle \phi \rangle$ and ϕ' , one can find a clear evidence that after control, much more heat is convected from the cylinder surface to the near-wall zone and the far wake. The high-temperature zones form two layers that corresponds to the two-row counter-rotating vortices, which is closely related to the vibration amplitude. Compared with the temperature field, the velocity fluctuations represented by u' and v' shows less differences in the controlled case, except for more concentrated fluctuations near the wall. The larger absolute values of $\langle u'\phi' \rangle$ and $\langle v'\phi' \rangle$ brings more insights that fluctuations of u and v cause positive influences on the fluctuation of temperature. Moreover, the influence of u' on ϕ' is evidenced both in the near-wall zone and in the far wake. By contrast, the influence of v' on ϕ' mainly occurs in the near-wall zone.

Finally, to quantify the contribution of the synthetic jets and the augmented VIV to the enhanced heat transfer, two extra experiments are further carried out. In the first experiment, the cylinder is fixed, whereas the synthetic jets issues exactly the identical forcing as the temporally varying U_{jet} shown in Fig. 7(c). The temporally varying U_{jet} is also shown in Fig. 12(a1). In the second experiment, the cylinder

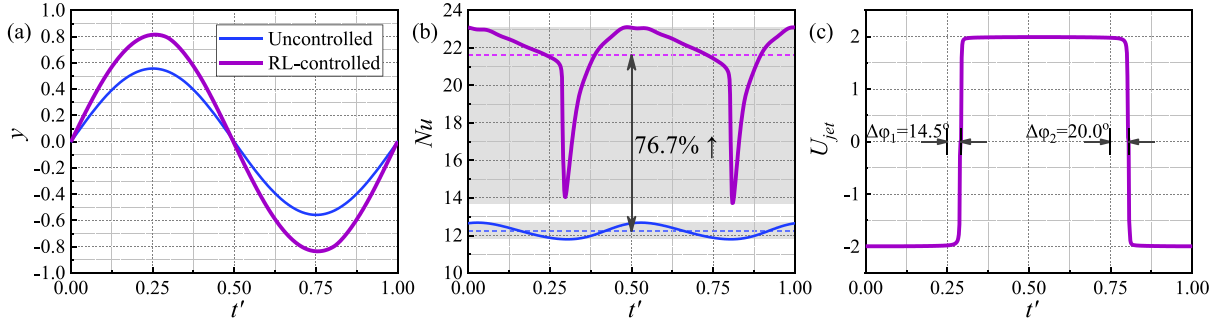


Fig. 8. Variations of (a) transverse displacement, (b) Nusselt number, and (c) velocity issued from the jet with the modified time t' in one vibration cycle.

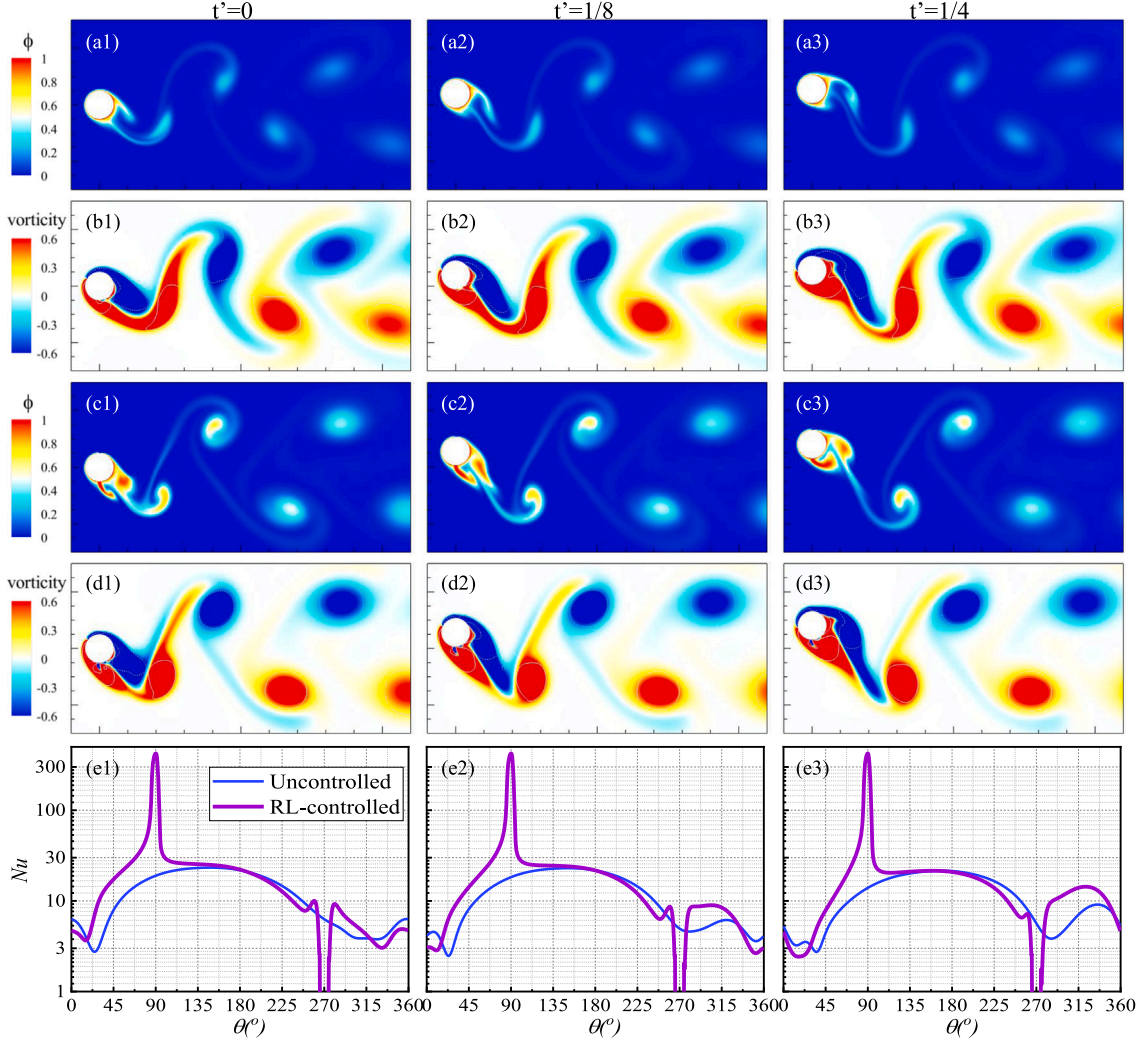


Fig. 9. Instantaneous snapshots of (a, c) temperature and (b, d) vorticity fields, together with (e) the instantaneous distribution of Nusselt number along the cylinder wall. Here (a–b) corresponds to the uncontrolled case at three instants. (c–d) correspond to the RL-controlled case.

does not issue any forcing, but is forced to vibrate with the temporally varying y shown in Fig. 7(b). The temporally varying y before and after control are also shown in Fig. 12(b1). Results of the two experiments are organized into Figs. 12(a) and 12(b), respectively. In Fig. 12(a), the mean Nusselt number in the quasi-steady state is 19.5, i.e., corresponding to 90.3% the Nusselt number of the RL-controlled VIV case. In Fig. 12(b), the solely augmented vibration only gains a 7.7% improvement of mean Nusselt number compared with the uncontrolled VIV case, which is far less than that of the first experiment. Note that due to the nonlinearity of the thermal-fluid-structure interaction system, the

whole gain ratio of Nu acquired in the two experiments is not equal to unity. Based on the above experiments and the quantitative evidences, it can be confirmed that the synthetic jets contributes significantly more than the augmented VIV.

3.3. Effect of the NOB approximation

With the optimal control strategy exploited in the last section, we proceed to consider a more realistic situation, where the fluid's kinematic viscosity and thermal diffusivity vary with temperature while

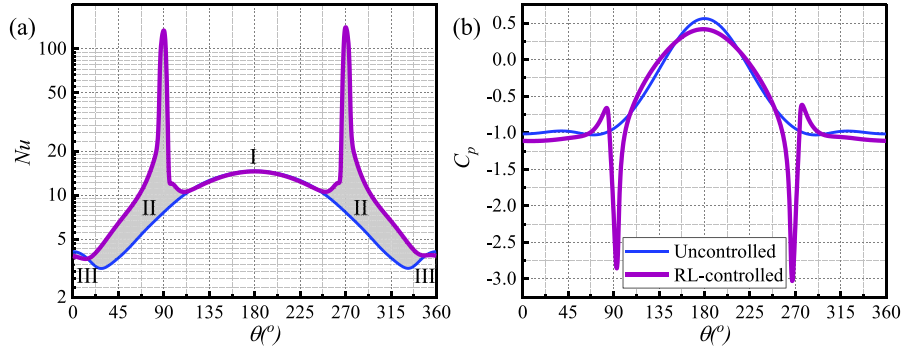


Fig. 10. Time-averaged (a) Nusselt number and (b) pressure coefficient distributed along the cylinder wall.

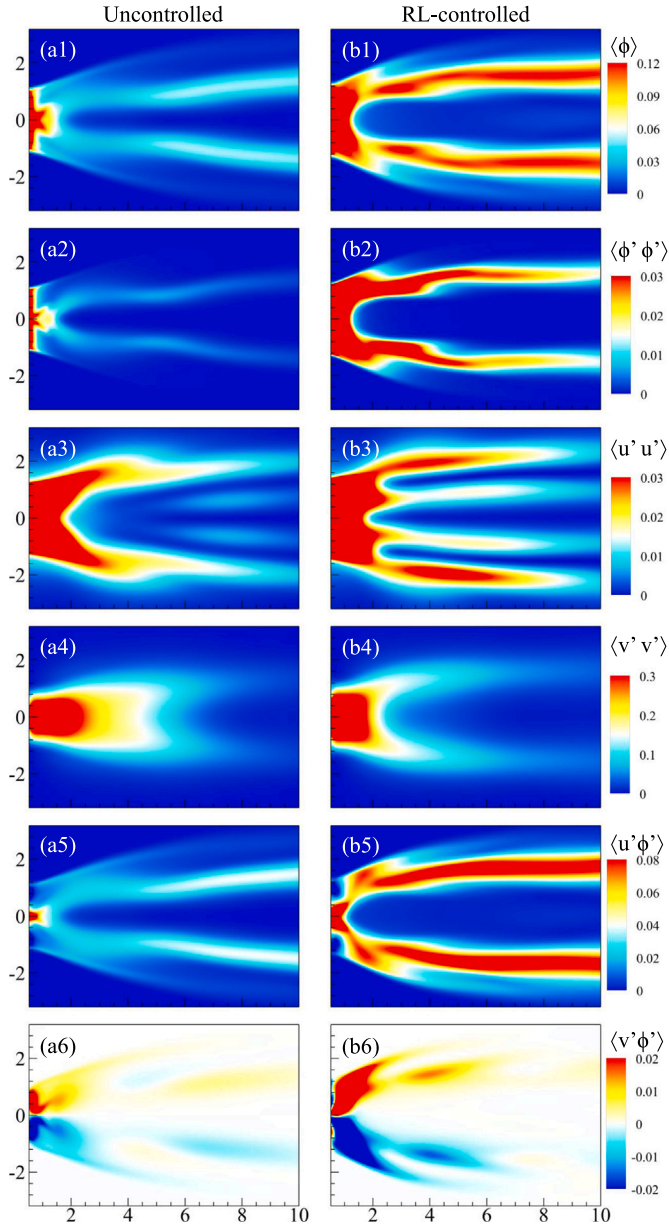


Fig. 11. Time averaged first-order and second-order statistical quantities.

the variation of density is neglected. In other words, the first type of non-Oberbeck–Bousinesq approximation (NOB-I) is adopted. The temperature difference between the cylinder surface and the environment is still fixed at 60 K. With the property equations provided by Zografos et al. [28], the relationship of kinematic viscosity as well as the Prandtl number against temperature as

$$\nu(T) = \frac{3.8208 \times 10^{-2}}{T - 252.33} (-3.0115 \times 10^{-6} T^3 + 9.6272 \times 10^{-4} T^2 - 0.11052T + 1022.4)^{-1} \quad (24)$$

$$Pr(T) = \frac{276.2}{T - 253.31} - 0.28661$$

The above relationships are also demonstrated in Fig. 13. When the temperature rises from 293.15 K to 353.15 K, the kinematic viscosity decreases by 58.4% while the Prandtl number decreases by 62.7%. Recalling the temperature distribution in Fig. 11, one can anticipate that more complexity would be brought to the thermal fluid–structure interaction environment if the NOB effect is taken into account. As stated in the work of Ali et al. [14], when the NOB effect is considered, the heat transfer induces a certain level of damping to the amplitude and frequency of the lift and drag.

In our simulations, the local kinematic viscosity and diffusivity are no longer fixed but determined by the local temperature according to Eq. (24). Thereafter the momentum and heat transport equation are coupled. Table 6 shows the mesh convergence test. A group of tests are carried out, where the environment temperature is fixed to be 273.15 K while the cylinder surface is heated to various values of temperature. The temperature difference ΔT is then set as 10 ~ 70 K, and one might expect that larger temperature difference leads to more significant of the NOB effect. For these ΔT 's, results are shown in Fig. 14. It is indicated that within the ΔT range discussed herein, characteristics regarding the heat transfer and structure vibration are significantly affected. It is unanticipated that both trends are not monotonic, for example, a peak Nu is observed when ΔT equals 60 K. This phenomenon might be resulted from the nonlinear nature of this thermal-fluid–structure interaction system.

In order to show the detailed influences and underlying physics of the NOB effect, the case with $\Delta T = 60$ K is then chosen as the representative case, whose near-wall distributions of temperature and vortical structures are then demonstrated in Fig. 15. In Fig. 15, the instantaneous snapshots are all selected when $y = 0$. At this instant, for the two uncontrolled cases, qualitatively only slight differences can be observed. However, the two controlled cases reveal quite different features. As can be noted from the quantitative results of Fig. 16, the jet velocities are both at its saturation state in the controlled cases, the lower jet issuing downward velocity. At the lower jet slot, the issued velocity in the NOB-controlled case leads to more complicated phenomenon. Comparing Fig. 15(d) with Fig. 15(b), one can observe an elongated clockwise vortex formed at the upper side, accompanied by a more evenly distributed high-temperature zone that can prompt heat transfer. Results in Fig. 16(b) confirm this judgment. In addition,

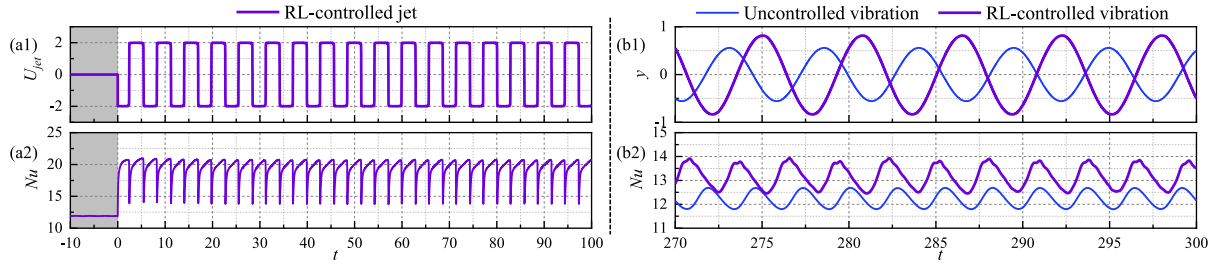


Fig. 12. Numerical experiment for quantifying contribution of synthetic jet and augmented vibration. (a) U_{jet} and Nu of the RL-controlled stationary cylinder. (b) y and Nu of the uncontrolled and RL-controlled vibrating cylinder.

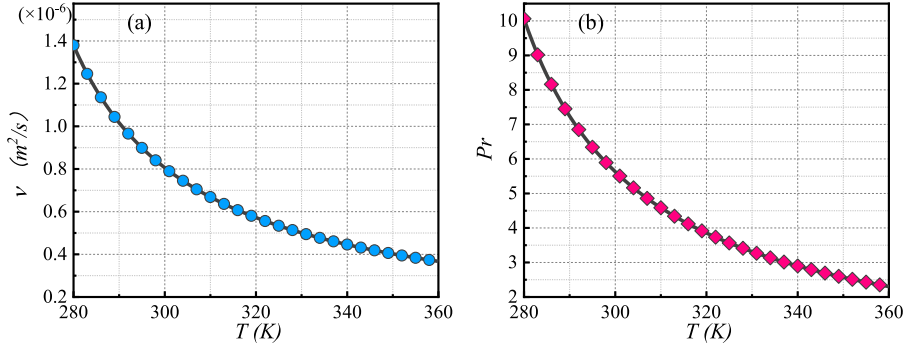


Fig. 13. Thermal properties of liquid water with temperature range 280 K–360 K.

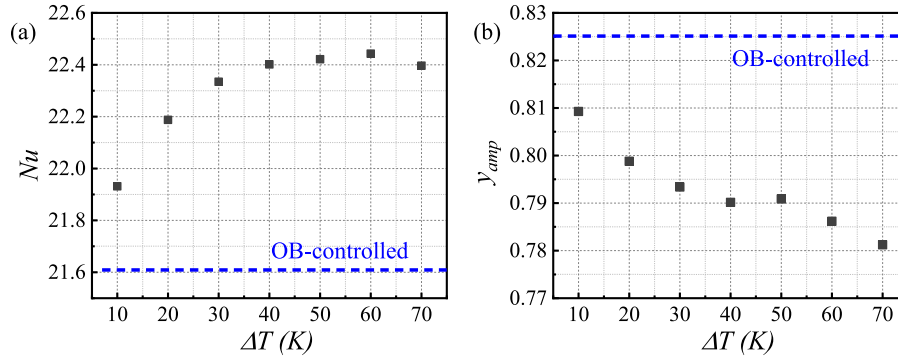


Fig. 14. Heat transfer rate and the VIV amplitude affected by different temperature differences due to the effect of the NOB effect.

differences of vorticity distribution in vicinity of the blowing jet sot can also be observed. Furthermore, it is noted that the vibration frequency in the NOB-controlled case is almost unaffected, the discrepancy with the OB-controlled case being only 0.5%.

In Fig. 16, the temporally varying data are once again arranged to one cycle, with the modified time. Compared with the OB-controlled case, the NOB-controlled case shows slightly smaller vibration amplitude but higher heat transfer rate, confirming again that the physical processes are quite different. Meanwhile, curves of U_{jet} almost coincides with each other, indicating that the control is almost unaffected by the NOB effect. Therefore, it can be confirmed through the above evidences that the control strategy is self-adaptive.

4. Summary

In this article, an intelligent flow control approach is proposed to enhance heat transfer from a circular cylinder at $Re = 100$, which is immersed in uniform upstream, elastically mounted with $m^* = 2$ and $U_R = 5$, and heated with constant temperature. In detail, our main contributions and findings are summarized as follows:

- The numerical solver established with body-fitted mesh and moving reference frame under the framework of the LBM provides a high-fidelity and computationally efficient tool for the present problem that couples flow, structural movement, and thermal convection.
- With sensorymotor cues providing the feedback signals, a pair of alternating blow/suction jets as the actuator, and the controller guided by the reinforcement learning, an optimal control strategy is achieved that successfully enhance the heat transfer rate by 76.7%.
- The mechanisms for the enhanced heat transfer is elucidated through analysis including the temporal varying flow and thermal quantities, statistical quantities, etc. The successful heat transfer enhancement is contributed mainly by the mediated vortex formation and shedding near the blowing jet, as well as the accompanying augmented vibration.

In this study, the closed-loop control system is simple yet efficient, and the RL is proven to be powerful in exploring the optimal control strategy. In the future, the present study would be extended to turbulent conditions which rely on high-fidelity three-dimensional simulations. In

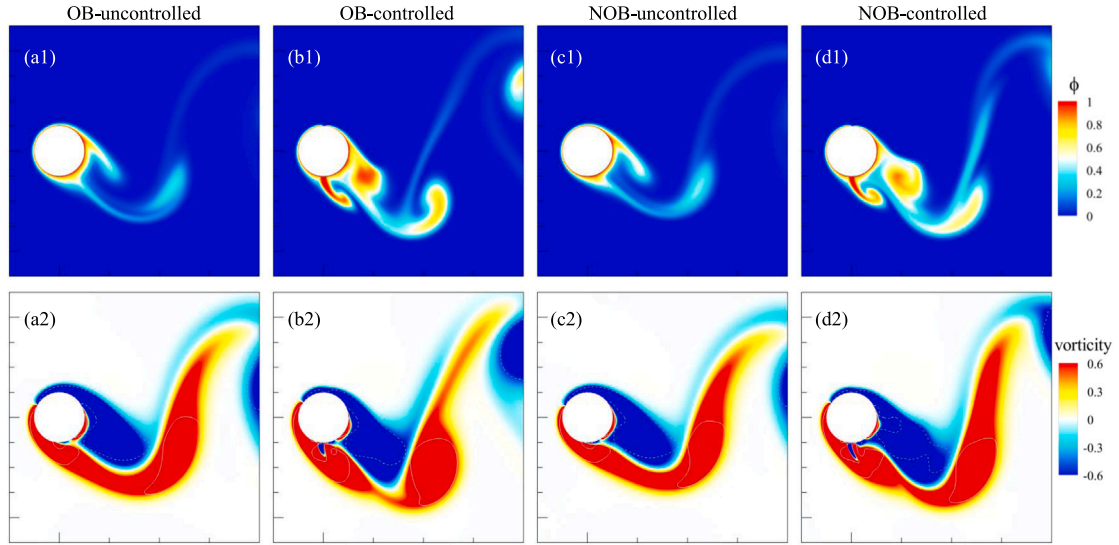


Fig. 15. Comparisons of instantaneous snapshots of near-wall temperature and vorticity when the cylinder is at its equilibrium position.

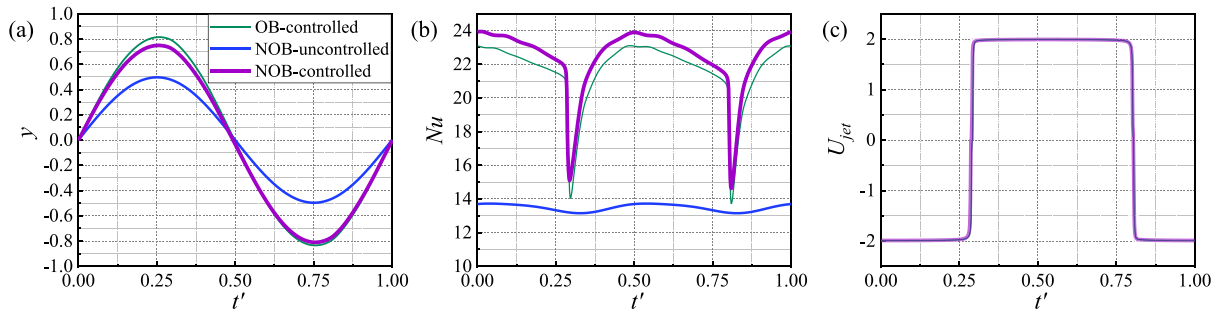


Fig. 16. Temporal variations of (a) transverse displacement, (b) Nusselt number, and (c) velocity issued from the jet with the NOB approximation.

addition, we anticipate that the present work would shed more lights to more complicated problems involving flow, structural motion, and heat transport.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

We gratefully acknowledge financial support from the National Natural Science Foundation of China (12102357), the Key Research and Development Program of Shaanxi (2023-YBGY-389), the Natural Science Foundation of Chongqing, China (cstc2021jcyj-msxmX0394), and the Fundamental Research Funds for the Central Universities (3102021HHZY030002).

Appendix

A.1. Validations of the numerical solver

As described detailedly in Section 2, our numerical solver combines the lattice Boltzmann particle collision process for both the momentum transport and the heat transport, the generalized interpolation-based implementation for particle streaming along the body-fitted O-type mesh, as well as the moving frame based scheme for body motions. In order to validate this complex solver, two scenarios that are related to the setups discussed in the main content are demonstrated. Detailed evidences and discussions are presented below. Note that in both cases, the Boussinesq approximation is applied, which is a more common practice.

The first scenario concerns the uniform flow past a stationary circular cylinder, where four Reynolds numbers are chosen, i.e., $Re = 100, 140, 150$, and 200 . Three Prandtl numbers are also chosen, i.e., $Pr = 0.7, 1$, and 10 . In this situation, global variables, including the drag coefficient C_D , the lift coefficient C_L , the Strouhal number S_r , and the Nusselt number Nu , are computed and organized into Table 2. To verify the mesh independence of our solver, we test three mesh resolutions, identical with those in Tables 5 and 6. Comparisons are made with prior published data [9,20,39,46,47]. As can be intuitively

Table 2
Validations and mesh independence test for the fixed cylinder case.

Re	Case	C_D	C_L	St	Nu_{mean}		
					$Pr = 0.7$	$Pr = 1$	$Pr = 10$
100	I (256 × 160)	1.281 ± 0.008	±0.306	0.160	4.998	5.646	10.319
	II (512 × 320)	1.327 ± 0.009	±0.325	0.164	5.120	5.830	12.992
	III (1024 × 640)	1.339 ± 0.010	±0.330	0.165	5.146	5.868	13.746
	Ren et al. [20]	1.375 ± 0.011	±0.341	0.169	...	...	...
	Zhou et al. [39]	1.307 ± 0.009	±0.324	0.162	...	5.84	13.61
	Patnana et al. [46]	...	...	...	...	5.85	13.94
	Mahir and Altaç [47]	1.368 ± 0.029	±0.343	0.172	5.179	...	...
	Izadpanah et al. [9]	1.373 ± 0.010	±0.339	0.169	5.159	...	...
	Sonawane et al. [7]	1.311 ± 0.0074	±0.308	0.167	5.118	...	...
140	I (256 × 160)	1.249 ± 0.019	±0.434	0.175	...	6.616	11.117
	II (512 × 320)	1.306 ± 0.022	±0.473	0.180	...	6.956	14.833
	III (1024 × 640)	1.322 ± 0.023	±0.484	0.181	...	7.022	16.239
	Zhou et al. [39]	...	...	...	6.22	6.95	16.07
	Patnana et al. [46]	...	...	...	...	7.03	16.58
150	I (256 × 160)	1.244 ± 0.022	±0.462	0.178	6.062	...	...
	II (512 × 320)	1.305 ± 0.025	±0.508	0.183	6.315	...	...
	III (1024 × 640)	1.323 ± 0.027	±0.521	0.184	6.366	...	...
	Zhou et al. [39]	1.283 ± 0.025	±0.500	0.181	...	...	...
	Izadpanah et al. [9]	1.355 ± 0.027	±0.534	0.187	6.395	...	...
	Sonawane et al. [7]	1.295 ± 0.0199	±0.506	0.186	6.308	...	...
	...	...	...	...	...	...	...
200	I (256 × 160)	1.233 ± 0.036	±0.582	0.189	6.900	...	...
	II (512 × 320)	1.309 ± 0.044	±0.663	0.195	7.323	...	...
	III (1024 × 640)	1.333 ± 0.046	±0.684	0.196	7.405	...	...
	Mahir and Altaç [47]	1.376 ± 0.048	±0.698	0.192	7.474	...	...
	Izadpanah et al. [9]	1.336 ± 0.046	±0.700	0.200	7.452	...	...
	Sonawane et al. [7]	1.302 ± 0.029	±0.6654	0.194	7.305	...	...
	...	...	...	...	...	...	...

Table 3

Validations and mesh independence test for the VIV cylinder case at $Re = 100$, $m^* = 2$, and $U_R = 5$.

Case	C_D	C_L	y_A	St
I (256 × 160)	2.077 ± 0.612	±0.547	±0.557	0.185
II (512 × 320)	2.095 ± 0.617	±0.576	±0.556	0.184
III (1024 × 640)	2.099 ± 0.618	±0.595	±0.557	0.183
Ren et al. [20]	...	...	±0.527	0.186
Wang et al. [48]	...	...	±0.545	0.186

observed from Table 2, for all quantities listed herein, the median mesh resolution (Mesh II) gives quite close results with those of the fine mesh resolution (Mesh III), and the discrepancy is within 5.5%. The coarse mesh resolution (Mesh I) gives the worst results that are not reliable. By comparison, results of the fine mesh resolution (Mesh III) reaches good agreement with previously data.

In the second scenario, the VIV is considered, and two parametric conditions are demonstrated below. In the first case, the Reynolds number is fixed at $Re = 100$, and the structural parameters are kept identical as those discussed in this article, i.e., $m^* = 2$ and $U_R = 5$. Due to very few available data concerning the thermal properties in this parametric condition, only the hydrodynamic forces and vibration characteristics are validated. Global variables, including the drag coefficient C_D , the lift coefficient C_L , the vibration amplitude y_A , and the Strouhal number St , are computed and organized into Table 3. Three mesh resolutions are also tested and show good convergence from Mesh I to Mesh III, where the discrepancy between Mesh II and Mesh III does not exceed 3.2%. The vibration amplitude and frequency are also consistent with prior data [20,48].

In the second case concerning the VIV, the Reynolds number is set as 150, the structural parameters are set as $m^* = 1.57$ and $U_R = 4$. The Prandtl number is fixed at $Pr = 7$, very close to the value considered in the main content. In addition to the hydrodynamic forces and vibration characteristics, the thermal Nusselt number is also computed and organized in Table 4. By comparison, it is confirmed that the present solver leads to very good agreement with the results of Izadpanah et al. [9] when dealing with thermal-fluid-structure interaction problems.

Table 4

Validations and mesh independence test for the VIV cylinder case at $Re = 150$, $m^* = 1.57$, $U_R = 4$, and $Pr = 7$.

Case	C_D	C_L	y_A	St	Nu
I (256 × 160)	2.133 ± 0.785	±1.355	±0.585	0.211	11.052
II (512 × 320)	2.133 ± 0.734	±1.431	±0.585	0.208	14.514
III (1024 × 640)	2.115 ± 0.732	±1.487	±0.586	0.207	15.593
Izadpanah et al. [9]	...	±1.429	±0.578	...	15.276

A.2. Mesh convergence of the present problem

The mesh convergence study is carried out in this section. Results are organized into Tables 5 and 6, corresponding to the OB approximation and the NOB approximation, respectively. The mesh convergence is examined by the grid convergence index (GCI) [49], and the results are organized into Table 7. Here the convergence order p_{co} is first calculated based on the mean Nusselt number of three mesh levels. It is then noted that the computed values are close to the value deduced via the Chapman-Enskog expansion, which show that the lattice Boltzmann method is of second-order accuracy in space. Based on this convergence order, the mean Nusselt number at zero mesh spacing is estimated via the Richardson extrapolation, which further clearly show that the numerical error of Nu_{mean} is only about 1% for both cases with the fine mesh (mesh III). Then the GCI can be computed. Here because three mesh levels are used in the mesh convergence tests, the safety factor 1.25 is adopted. The GCI for grid I and grid II, as well as the GCI for grid II and grid III, are then illustrated in Table 7. The very small values of $GCI_{I,II}$ confirms that the mesh is well converged when the fine mesh is used. Moreover, for both cases, the relationship $GCI_{I,II}/GCI_{I,II} \approx 2^{p_{co}}$ implies that the solutions are in the asymptotic range of convergence.

Considering the fact that the median mesh provides comparable results with the fine mesh, while the computational cost of the median mesh setup can be reduced to a quite large extent, we adopt the median mesh (mesh II) in the training process. In the deterministic run, we turn to the fine mesh (mesh III) to provide more accurate data. This practice has been verified to be adequate for machine learning based active flow control [22].

Table 5

Mesh independence test for the Oberbeck–Bousinesq approximation.

Case	C_D	C_L	y_A	Nu_{min}	Nu_{max}	Nu_{mean}	St
I (256 × 160)	2.077 ± 0.612	±0.547	±0.557	9.554	10.111	9.833	0.185
II (512 × 320)	2.095 ± 0.617	±0.576	±0.556	11.390	12.219	11.805	0.184
III (1024 × 640)	2.099 ± 0.618	±0.595	±0.557	11.797	12.689	12.243	0.183

Table 6

Mesh independence test for the non-Bousinesq approximation.

Case	C_D	C_L	y_A	Nu_{min}	Nu_{max}	Nu_{mean}	St
I (256 × 160)	2.128 ± 0.589	±0.348	±0.513	10.334	10.690	10.512	0.190
II (512 × 320)	2.222 ± 0.583	±0.383	±0.502	12.658	13.188	12.923	0.188
III (1024 × 640)	2.279 ± 0.581	±0.407	±0.497	13.148	13.726	13.437	0.187

Table 7

Examining mesh convergence using grid convergence index.

Case	Convergence order p_{co}	$Nu_{mean,0}$	$GCI_{I,II}$ (%)	$GCI_{II,III}$ (%)
OB	2.169	12.368	5.971	1.280
NOB	2.229	13.576	6.319	1.296

References

- [1] J.F. Derakhshandeh, M.M. Alam, A review of bluff body wakes, *Ocean Eng.* 182 (2019) 475–488.
- [2] H. Choi, W.-P. Jeon, J. Kim, Control of flow over a bluff body, *Annu. Rev. Fluid Mech.* 40 (2008) 113–139.
- [3] S.S. Collis, R.D. Joslin, A. Seifert, V. Theofilis, Issues in active flow control: Theory, control, simulation, and experiment, *Prog. Aerosp. Sci.* 40 (2004) 237–289.
- [4] A. Arshad, M. Jabbar, Y. Yan, Synthetic jet actuators for heat transfer enhancement—A critical review, *Int. J. Heat Mass Transfer* 146 (2020) 118815.
- [5] C.H.K. Williamson, R. Govardhan, Vortex-induced vibrations, *Annu. Rev. Fluid Mech.* 36 (2004) 413–455.
- [6] S. Bao, S. Chen, Z. Liu, C. Zheng, Lattice Boltzmann simulation of the convective heat transfer from a stream-wise oscillating circular cylinder, *Int. J. Heat Fluid Flow* 37 (2012) 147–153.
- [7] C. Sonawane, P. Praharaj, A. Kulkarni, A. Pandey, H. Panchal, Numerical simulation of heat transfer characteristics of circular cylinder forced to oscillate elliptically in an incompressible fluid flow, *J. Therm. Anal. Calorim.* 148 (2023) 2719–2736.
- [8] Q.M. Al-Mdallal, F.M. Mahfouz, Heat transfer from a heated non-rotating cylinder performing circular motion in a uniform stream, *Int. J. Heat Mass Transfer* 112 (2017) 147–157.
- [9] E. Izadpanah, Y. Amini, A. Ashouri, A comprehensive investigation of vortex induced vibration effects on the heat transfer from a circular cylinder, *Int. J. Therm. Sci.* 125 (2018) 405–418.
- [10] C. Sonawane, P. Praharaj, A. Pandey, A. Kulkarni, K. Kotecha, H. Panchal, Case studies on simulations of flow-induced vibrations of a cooled circular cylinder: Incompressible flow solver for moving mesh problem, *Case Stud. Therm. Eng.* 34 (2022) 102030, <http://dx.doi.org/10.1016/j.csite.2022.102030>, URL: <https://www.sciencedirect.com/science/article/pii/S2214157X22002763>.
- [11] L. Ding, Y. Han, Z. Yang, L. Zhang, H. He, Influence of upstream cylinder on flow-induced vibration and heat transfer of downstream cylinder, *Int. J. Therm. Sci.* 176 (2022) 107519.
- [12] X. Sun, S. Li, G.-G. Lin, J.-Z. Zhang, Effects of flow-induced vibration on forced convection heat transfer from two tandem circular cylinders in laminar flow, *Int. J. Mech. Sci.* 195 (2021) 106238.
- [13] V. Kumar, H. Garg, G. Sharma, R. Bhardwaj, Harnessing flow-induced vibration of a D-section cylinder for convective heat transfer augmentation in laminar channel flow, *Phys. Fluids* 32 (2020) 083603.
- [14] U. Ali, M.D. Islam, I. Janajreh, Flow over rotationally oscillating heated circular cylinder at low Reynolds number, *Ocean Eng.* 265 (2022) 112515.
- [15] C. Ozalp, D.B. Saydam, C. Polat, M. Soyler, E. Hürdoğan, The effect of a high-speed rotational oscillating cylinder on a heated cylinder, *Int. J. Heat Fluid Flow* 95 (2022) 108971.
- [16] S.L. Brunton, B.R. Noack, P. Koumoutsakos, Machine learning for fluid mechanics, *Annu. Rev. Fluid Mech.* 52 (2020) 477–508.
- [17] F. Ren, H. Hu, H. Tang, Active flow control using machine learning: A brief review, *J. Hydrodyn.* 32 (2020) 247–253.
- [18] J. Rabault, F. Ren, W. Zhang, H. Tang, H. Xu, Deep reinforcement learning in fluid mechanics: A promising method for both active flow control and shape optimization, *J. Hydrodyn.* 32 (2020) 234–246.
- [19] C. Vignon, J. Rabault, R. Vinuesa, Recent advances in applying deep reinforcement learning for flow control: Perspectives and future directions, *Phys. Fluids* 35 (2023) 031301.

- [20] F. Ren, C. Wang, H. Tang, Active control of vortex-induced vibration of a circular cylinder using machine learning, *Phys. Fluids* 31 (2019) 093601.
- [21] J. Rabault, M. Kuchta, A. Jensen, U. Réglade, N. Cerardi, Artificial neural networks trained through deep reinforcement learning discover control strategies for active flow control, *J. Fluid Mech.* 865 (2019) 281–302.
- [22] F. Ren, J. Rabault, H. Tang, Applying deep reinforcement learning to active flow control in weakly turbulent conditions, *Phys. Fluids* 33 (2021) 037121.
- [23] P. Varela, P. Suárez, F. Alcántara-Ávila, A. Miró, J. Rabault, B. Font, L.M. García-Cuevas, O. Lehmkuhl, R. Vinuesa, Deep reinforcement learning for flow control exploits different physics for increasing Reynolds number regimes, *Actuators* 11 (2022) 359.
- [24] F. Ren, C. Wang, H. Tang, Bluff body uses deep-reinforcement-learning trained active flow control to achieve hydrodynamic stealth, *Phys. Fluids* 33 (2021) 093602.
- [25] C. Zheng, T. Ji, F. Xie, X. Zhang, H. Zheng, Y. Zheng, From active learning to deep reinforcement learning: Intelligent active flow control in suppressing vortex-induced vibration, *Phys. Fluids* 33 (2021) 063607.
- [26] G. Beintema, A. Corbetta, L. Biferale, F. Toschi, Controlling Rayleigh–Bénard convection via reinforcement learning, *J. Turbul.* 21 (2020) 585–605.
- [27] L. Guastoni, J. Rabault, P. Schlatter, H. Azizpour, R. Vinuesa, Deep reinforcement learning for turbulent drag reduction in channel flows, *Eur. Phys. J. E* 46 (2023) 27.
- [28] A.I. Zografos, W.A. Martin, J.E. Sunderland, Equations of properties as a function of temperature for seven fluids, *Comput. Methods Appl. Mech. Engrg.* 61 (1987) 177–187.
- [29] L. Li, S.J. Sherwin, P.W. Bearman, A moving frame of reference algorithm for fluid/structure interaction of rotating and translating bodies, *Internat. J. Numer. Methods Fluids* 38 (2002) 187–206.
- [30] H. Huang, M. Sukop, X. Lu, Multiphase Lattice Boltzmann Methods: Theory and Application, John Wiley & Sons, 2015.
- [31] D. d’Humières, Multiple-relaxation-time lattice Boltzmann models in three dimensions, *Phil. Trans. R. Soc. A* 360 (2002) 437–451.
- [32] X. He, L.-S. Luo, Lattice Boltzmann model for the incompressible Navier–Stokes equation, *J. Stat. Phys.* 88 (1997) 927–944.
- [33] Z. Guo, C. Zheng, B. Shi, Discrete lattice effects on the forcing term in the lattice Boltzmann method, *Phys. Rev. E* 65 (2002) 046308.
- [34] F. Ren, B. Song, H. Hu, Lattice Boltzmann simulations of turbulent channel flow and heat transport by incorporating the vremen model, *Appl. Therm. Eng.* 129 (2018) 463–471.
- [35] F. Ren, B. Song, Y. Zhang, H. Hu, A GPU-accelerated solver for turbulent flow and scalar transport based on the lattice Boltzmann method, *Comput. & Fluids* 173 (2018) 29–36.
- [36] L. Li, R. Mei, J.F. Klausner, Lattice Boltzmann models for the convection-diffusion equation: D2Q5 vs D2Q9, *Int. J. Heat Mass Transfer* 108 (2017) 41–62.
- [37] L. Jahanshaloo, N.A.C. Sidik, A. Fazeli, M.P. HA, An overview of boundary implementation in lattice Boltzmann method for computational heat and mass transfer, *Int. Commun. Heat Mass Transfer* 78 (2016) 1–12.
- [38] T. Imamura, K. Suzuki, T. Nakamura, M. Yoshida, Acceleration of steady-state lattice Boltzmann simulations on non-uniform mesh using local time step method, *J. Comput. Phys.* 202 (2005) 645–663.
- [39] X. Zhou, B. Dong, C. Chen, W. Li, A thermal LBM-LES model in body-fitted coordinates: Flow and heat transfer around a circular cylinder in a wide Reynolds number range, *Int. J. Heat Fluid Flow* 77 (2019) 113–121.
- [40] X. Zhou, B. Dong, W. Li, C. Chen, A 3D thermal LB model on non-orthogonal grid and its application for natural convection in irregular domains, *Int. J. Heat Mass Transfer* 136 (2019) 223–234.
- [41] Z.-L.G. Guo, C.-G. zheng, B.-C. Shi, Non-equilibrium extrapolation method for velocity and pressure boundary conditions in the lattice Boltzmann method, *Chin. Phys. B* 11 (2002) 366.
- [42] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, 2017, arXiv:1707.06347.
- [43] P.-W. Chou, D. Maturana, S. Scherer, Improving stochastic policy gradients in continuous control with deep reinforcement learning using the beta distribution, in: *Proceedings of the 34th International Conference on Machine Learning*, vol. 70, 2017, pp. 834–843.
- [44] D.P. Kingma, J. Ba, Adam: A method for stochastic optimization, 2014, arXiv: 1412.6980.
- [45] J. Zhou, R.J. Adrian, S. Balachandar, T.M. Kendall, Mechanisms for generating coherent packets of hairpin vortices in channel flow, *J. Fluid Mech.* 387 (1999) 353–396.
- [46] V.K. Patmana, R.P. Bharti, R.P. Chhabra, Two-dimensional unsteady forced convection heat transfer in power-law fluids from a cylinder, *Int. J. Heat Mass Transfer* 53 (2010) 4152–4167.
- [47] N. Mahir, Z. Altaç, Numerical investigation of convective heat transfer in unsteady flow past two cylinders in tandem arrangements, *Int. J. Heat Fluid Flow* 29 (2008) 1309–1318.
- [48] C. Wang, H. Tang, F. Duan, C.M. Simon, Control of wakes and vortex-induced vibrations of a single circular cylinder using synthetic jets, *J. Fluids Struct.* 60 (2016) 160–179.
- [49] P.J. Roache, Perspective: A method for uniform reporting of grid refinement studies, *J. Fluids Eng.* 116 (1994) 405–413, <http://dx.doi.org/10.1115/1.2910291>.